Amna Zia

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LinkedIn
Portfolio

Education

- **Georgia Institute of Technology** — Atlanta, GA
  Bachelor of Science in Mechanical Engineering — *Expected May 2028*
  - Dean's List
  - Coursework: Statics, MATLAB, Linear Algebra, Engineering Graphics, Engineering Materials, Mechanical Design

Experience

- **Data Analytics Externship** — Remote
  NASCAR, Extern — *June 2026 – July 2026*
  - Analyzed NASCAR race and sponsorship data using Python, APIs, and web scraping to identify sponsor visibility trends.
- **AI/ML Research Assistant** — Atlanta, GA
  Flow Physics and Computational Science Lab, Georgia Tech — *Dec 2025 – Present*
  - Generated high-fidelity 3D multiphase turbulence simulations on Aurora, contributing computational data to MFlowDB, an open-source dataset developed with Argonne National Laboratory and NVIDIA.
  - Configured and validated simulations using fluid mechanics principles to support physically consistent simulation outputs.
  - Benchmarked deep learning models (autoencoders, neural operators) for predicting flow evolution across turbulence regimes, applied SHAP to identify influential flow features and improve model interpretability.
- **FRC Robotics**
  Lead Team Member — *Sep 2021 – May 2024*
  - Led the design and development of competition robots, integrating mechanical subsystems with programming and control systems.
  - Mentored 10+ team members in programming and design, developing technical skills and supporting collaborative project execution.

Projects

- **Autonomous Arena Mechatronic System – Mechanical Design Project** — Atlanta, GA
  Mechanical Design Course, Georgia Institute of Technology — *Aug 2026 – Present*
  - Designed and assembled an autonomous robot featuring a motor-driven drivetrain, pneumatic flapper mechanism, and solenoid-actuated release door.
  - Fabricated and assembled mechanical components using hand tools, power drills, and shop equipment to support prototype development.
  - Programmed a Teensy 4.1 microcontroller in C to coordinate electromechanical mechanisms.
- **Row-Weeding Robot – Drivetrain CAD Design** — Atlanta, GA
  Mechanical Design Project, Georgia Institute of Technology — *Jan 2025 – May 2025*
  - Modeled a multi-part wheel and drivetrain subsystem for an autonomous agricultural robot using SolidWorks, developing components, assemblies, and motion constraints.
  - Improved fit, stability, and manufacturability through iterative CAD, tolerance analysis, and chassis integration.
- **Crescent Lantern – Product Design Project** — Atlanta, GA
  3D Mechanical Design Project, Georgia Institute of Technology — *Jan 2025 – May 2025*
  - Designed a multifunctional lantern in SolidWorks and developed 3D-printed prototypes to evaluate component fit, functionality, and assembly.
  - Iterated on CAD geometry and physical prototypes to improve manufacturability and product functionality.

Skills

- **CAD & Mechanical Design**: SolidWorks (Parts, Assemblies, Engineering Drawings), GD&T, Fits & Clearances, Tolerance Analysis, DFM/DFA, Bill of Materials (BOM)
- **Prototyping & Fabrication**: 3D Printing, Laser Cutting, Hand Tools, Power Drills, Band Saw Operation, Hands-on Assembly, Iterative Prototyping
- **Mechanical Systems**: Drivetrain Design, Motor-Driven Mechanisms, Pneumatics, Solenoid Actuation, Mechanical Integration
- **Programming & Controls**: MATLAB, Python, C, Microcontrollers (Teensy 4.1)
- **Software**: Microsoft Excel, Word, PowerPoint, Outlook

Activities

- **Society of Women Engineers (SWE), Georgia Tech**
  Member — *Aug 2024 – Present*
- **Junior STEM, Georgia Tech**
  Volunteering Member — *Jan 2025 – Present*
- **HyTech Racing, Georgia Tech**
  Mechanical Engineering Team — *Aug 2026 – Present*